# Could Artificial Intelligence Be Replaced With Biocomputing Intelligence?

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**Abstract —** Biocomputing systems built from living neurons—most visibly Cortical Labs' DishBrain and CL1, which taught cultured neurons to play Pong and, more recently, Doom—are increasingly framed as a resource-driven successor to conventional silicon AI. This paper evaluates that claim against three necessary conditions: capability parity, resource superiority at data-center scale, and the absence of a disqualifying hidden cost, mapping three existing systems onto the evaluation. On capability, biocomputing systems show real, measured learning, but independent testing on one system found most of the apparent intelligence traced to conventional silicon scaffolding, not the biological substrate—parity is partial and unverified at the substrate level. On resource superiority, the efficiency case is real per neuron but has no demonstrated path to scale: without a vascular system, cultured tissue cannot sustain a viable core beyond roughly 475 micrometers in radius, orders of magnitude below data-center scale, and no system has published a solution. On hidden cost, capability and moral-status risk are logically independent properties, not two ends of one dial—shown directly by a 2022 paper's use of "sentience" for a capability finding and the 30-scientist rebuttal it triggered—so every efficiency gain biocomputing offers is bought with a risk category silicon AI does not carry. The honest current answer is no, ordered by which gap is most urgent to close: not because the biology is uninteresting, but because condition two fails on a specific, peer-reviewed physical constraint funding and algorithms cannot resolve, and condition three imposes a cost no public comparison currently prices.

**Index Terms** — biocomputing, organoid intelligence, artificial intelligence comparison, moral status, consciousness science, risk stratification

## I. Introduction

In March 2025, Cortical Labs began shipping a $35,000 desktop unit built around living human neurons [1]. The unit, called CL1, is a commercial descendant of DishBrain, a 2022 research prototype in which roughly 800,000 cultured neurons learned to play Pong within minutes of real-time gameplay [2]. In 2026, an independent collaborator used Cortical Labs' own API to get a CL1 unit playing Doom—a substantially harder, non-linear, exploration-requiring task—in under a week, a detail the company narrates directly in its own product demonstration [3]. Both results have been covered widely in general and technology media, in registers ranging from straightforward technical explanation [4] to explicit alarm about whether the technology raises questions society is not equipped to answer [5], and a consistent thread runs through nearly all of that coverage: biocomputing is framed as a response to conventional AI's resource crisis, a biological substrate that could someday replace silicon because it learns from less data on a fraction of the power. None of that popular coverage is treated in this paper as scientific evidence; we cite it only because it accurately reflects the terms in which this technology is actually being discussed outside the laboratory—and because that framing deserves a direct, falsifiable answer rather than a vague sense that the technology "could eventually" scale.

This paper supplies that answer by decomposing the replacement claim into three conditions that all have to hold simultaneously, not just individually plausible in isolation. *Capability parity* asks whether the biological substrate itself, not surrounding conventional hardware, is doing the learning. *Resource superiority at scale* asks whether the efficiency case that motivates the whole comparison actually has a demonstrated path from a single cultured dish to anything resembling data-center throughput. *No disqualifying cost* asks whether pursuing that efficiency case is free of a cost silicon AI does not carry at all. A "yes" to the paper's title question requires all three. This paper finds one holds only partially, one fails on a specific physical constraint, and one fails by construction—and argues that finding is more useful than an aggregate verdict, because each condition points to a different, independently addressable research gap.

The capability question has a precise and instructive origin. When Kagan et al. published the DishBrain result, they described the neuron culture as exhibiting "sentience"—defined in the paper in a narrow, functional sense: responsiveness to sensory input in a goal-directed way [2]. Thirty neuroscientists signed a published rebuttal whose primary and most developed concern was exactly that terminology: that "sentience," "intelligence," and "goal-directed behavior" were being applied to cultured neurons without the conceptual grounding those terms carry elsewhere in neuroscience and bioethics, and that this risked misleading both scientific peers and the public about what the result actually showed—including, the signatories noted, creating a downstream financial incentive to keep using inflated language for the methods described [6]. But the

rebuttal was not purely terminological: it also stated directly that some of the study's conclusions were compromised by results that failed to adequately match control and experimental conditions—a methodological objection that brushes against the capability claim itself, not only its label. Kagan et al.'s own published reply concedes the terminology critique is worth taking seriously while defending their operational definition as principled rather than promotional, and does not substantively rebut the control-condition concern [7]. This paper treats that dispute as the first documented instance of exactly the pattern a replacement claim risks repeating at larger scale: a capability finding gets described in language that implies more than the data supports, and the field has not adopted a vocabulary that would prevent the same overreach from happening again with the next milestone—including, this paper argues, the milestone of "ready to replace conventional AI."

The contributions of this paper are as follows:

- We decompose the AI-replacement claim into three explicit, individually testable necessary conditions—capability parity, resource superiority at scale, and no disqualifying cost—rather than evaluating it as one aggregate question, so that each finding points to a specific, addressable gap rather than a diffuse verdict.
- On resource superiority, we identify a concrete bioengineering limit—the diffusion-limited size ceiling on unvascularized biological tissue—that caps viable biocomputing substrate far below what data-center-scale deployment would require, and that no biocomputing system studied here has published a solution to it.
- On disqualifying cost, we give a construction argument, not an empirical one, for why a biocomputing system's demonstrated capability and its moral-status risk can vary on separate axes entirely rather than trading off along one—a claim that does not require solving the hard problem of consciousness to be useful, only requiring that capability and risk not be conflated in language or governance.
- We map three existing systems—DishBrain, CL1, and FinalSpark's Neuroplatform—against all three conditions, and compare biocomputing to conventional silicon AI across the specific dimensions the conditions require: energy efficiency, data efficiency, scale trajectory, and the risk dimension that comparison usually omits entirely.

The remainder of this paper proceeds as follows. Section II reviews three literatures this paper draws on: biocomputing and organoid intelligence, consciousness science, and AI capability evaluation. Section III traces the technical lineage from DishBrain to CL1 and the training mechanism both rely on. Section IV evaluates the three necessary conditions in turn, including the capability/risk framework and its theoretical grounding. Section V maps three systems onto that evaluation. Section VI states limitations directly. Section VII concludes with a direct answer and a research agenda ordered by urgency.

## II. Related Work

### A. Biocomputing and Organoid Intelligence

The technical foundation for growing neurons that were never part of a living brain traces to Yamanaka's reprogramming work: four defined transcription factors—Oct3/4, Sox2, Klf4, and c-Myc—can reset a mature skin cell back to a pluripotent state from which it can be redirected into almost any cell type, including a neuron [8]. The human-cell version of this result, published a year later, meant researchers no longer needed neural tissue extracted from a living or donated brain; a small skin sample was sufficient [9]. Every biocomputing system discussed in this paper depends on this result, directly or through cultures descended from it. Kagan et al.'s DishBrain assembled roughly 800,000 such neurons on a high-density multielectrode array, embedded the culture in a closed-loop simulated Pong game-world, and reported a statistically significant increase in successful paddle returns within five minutes of gameplay—the paper's operational definition of learning [2]. A large multi-author position paper coined the term "organoid intelligence" for the broader research program this result opened, arguing that three-dimensional neural organoids—rather than flat cultures—could eventually support far more complex computation, and grounding the case for pursuing this on efficiency terms: an estimated $10^6$-fold power advantage over a comparable digital system on certain classes of problem, and comparable data efficiency claims relative to conventional machine learning [10]. FinalSpark's Neuroplatform operationalizes a related but distinct model: sixteen brain organoids, roughly 160,000 total neurons, made remotely accessible online for a monthly access fee, described in a peer-reviewed system paper that documents the platform's architecture without making the more dramatic efficiency claims that have since circulated in press coverage about the company [11]. None of this literature evaluates the replacement question against explicit necessary conditions; each paper argues its own system is a promising step, not that biocomputing as a category currently satisfies what replacing conventional AI would require. That is the gap this paper works to fill.

### B. Consciousness Science and Moral Status

Chalmers' original statement of the hard problem draws the distinction Section IV's third condition depends on: explaining a system's functional, behavioral capacities is one problem, and explaining why any of that functioning is accompanied by subjective experience at all is a separate, harder problem that functional explanation does not automatically resolve [12]. Solms argues for a specific, falsifiable-in-principle position on where experience originates: not as a property that emerges once cognition becomes sufficiently sophisticated, but as more evolutionarily basic than cognition, arising from homeostatic, affect-regulating circuitry that predates cortex [13]. This matters directly for biocomputing, because the free-energy-principle training method every system in this paper's case mapping uses—rewarding a culture with predictable stimulation when it succeeds and punishing it with chaotic, unpredictable stimulation when it fails—is architecturally a ver-

sion of exactly the predict-and-minimize-surprise loop Solms locates as affect's origin [2]. If Solms is right, that training method is not incidental to the cost question this paper's third condition asks about; it is that question, applied as an engineering technique rather than examined as one. Independently of any theory of where experience originates, there is direct empirical evidence that humans are poor at detecting its presence even in other humans. Owen et al. found that a patient who met every clinical criterion for a vegetative diagnosis showed fMRI activation, on request to imagine playing tennis, indistinguishable from healthy volunteers performing the same task—evidence of covert awareness a bedside behavioral exam had missed entirely [14]. Two independent follow-up studies converge on this not being a rare exception: Monti et al. found four of twenty-three behaviorally vegetative patients could willfully modulate their own brain activity on command [15], and a larger, more recent multi-site study found evidence of this covert cognitive-motor dissociation in approximately one quarter of clinically unresponsive patients [16], while a systematic review using more conservative criteria places the figure closer to fifteen percent [17]. The exact rate is contested; the direction of the finding is not. If clinicians with a fully human patient in front of them, using validated diagnostic criteria, miss covert awareness in roughly one in six to one in four cases, the base rate for correctly assessing whatever is or is not present in a dish of neurons should be assumed worse, not better, absent a specific reason to think otherwise.

### C. AI Capability Evaluation

A separate, purely computational literature establishes the terms this paper's resource-superiority condition needs. Patterson et al. measured GPT-3's training energy consumption directly at 1,287 megawatt-hours, with total carbon footprint varying by up to a factor of fifty depending on the training data center's local grid composition [18]—a precise, audited figure this paper prefers over the rounder, less-sourced estimates that circulate in secondary biocomputing literature. Epoch AI's continuously updated compute-trend analysis finds that training compute for notable AI models has grown roughly 4.4-fold per year since 2010, a pace that has further accelerated for frontier models released since 2020, with the most recent releases showing doubling times as short as three to four months [19]. Neither of these sources is biocomputing literature; we use them because a fair replacement evaluation requires a real, audited baseline for what conventional AI actually costs, not a figure borrowed secondhand from a biocomputing paper's own citation of it. Underlying both the biocomputing efficiency claims and the consciousness-science affect argument is a basic physiological fact: the human brain, at roughly two percent of body mass, consumes approximately twenty percent of resting metabolic energy, translating to an estimated twenty watts of continuous power draw [20], [21]. This number appears, uncredited to its physiological source, in nearly every popular account of biocomputing's efficiency case; we cite the primary neuroscience literature directly rather than repeat it as an unattributed round number.

## III. Background: From DishBrain to CL1

Both systems this paper treats as primary cases share one training mechanism, first demonstrated at scale by DishBrain and carried forward, apparently unchanged in its basic logic, into CL1. A living neuron culture sits on a multielectrode array divided into sensory and motor regions. A game engine translates its internal state—ball position in Pong, an approaching enemy in Doom—into an electrical stimulation pattern delivered to the sensory region. The culture's spontaneous and evoked firing in the motor region is read back out and mapped to an action: paddle up or down; move, turn, or shoot. Neither game engine grants the culture any innate knowledge of what it is controlling. The behavioral shaping happens entirely through consequence: a missed action is followed by unpredictable, high-entropy stimulation across the array, and a successful action is followed by calm, predictable stimulation [2]. Kagan and colleagues explain this design choice by appeal to the free energy principle—the proposition that living systems generally act so as to minimize surprise, because an organism that cannot predict the consequences of its own actions cannot reliably keep itself alive. Under that account, the culture is not learning to win a game in any sense that requires understanding what a game is; it is minimizing an aversive signal, and winning the game is the observable side effect.

Cortical Labs' own account of moving this mechanism from Pong to Doom is instructive about what, specifically, improved between 2022 and 2026—and it is the paper's first evidence relevant to the capability-parity condition. Pong offers a single, direct input-output relationship—ball position maps to one correct paddle direction—and took the company's original hardware and software over eighteen months to get working reliably. Doom is qualitatively harder: three-dimensional, populated with enemies in varying positions, requiring the system to explore an environment rather than track one variable. Using a matured software interface the company calls its API, running on a cloud platform it calls Cortical Cloud, an independent collaborator implemented the Doom mapping—enemy position to directional stimulation, spike pattern to shoot-or-turn-or-move command—in under a week [3]. The company is explicit in its own account that the resulting player is not skilled: it dies constantly, plays, in the presenting scientist's own phrase, "like a beginner who's never seen a computer," but shows consistent goal-seeking behavior—locating and firing at enemies—that counts as evidence of ongoing learning [3]. What changed across four years was the interface and the encoding scheme, not the underlying biological mechanism, which is the same predict-and-minimize-surprise loop DishBrain used. As Section IV-A discusses, that distinction matters directly for capability parity: an improved interface is evidence the surrounding engineering matured, not by itself evidence the biological substrate's own contribution grew at the same rate.

Fig. 1 depicts this shared mechanism. One inconsistency is worth naming directly rather than silently resolving, because it illustrates the precision problem this paper argues the field has more broadly whenever a capability or scale claim is

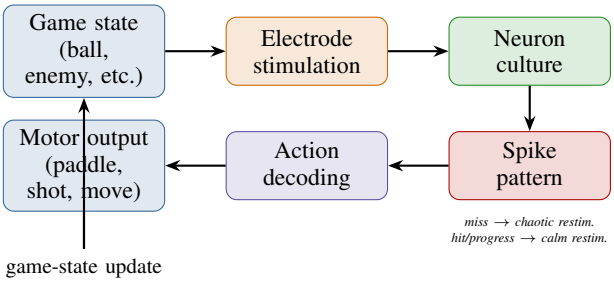



Fig. 1. The closed-loop training mechanism shared by DishBrain and CL1: game state is encoded as electrode stimulation, the neuron culture's spiking response is decoded into a motor action, and the resulting outcome determines whether the next stimulation is calm/predictable (success) or chaotic/unpredictable (failure) [2], [3].

made. Cortical Labs' own launch-period press materials and independent technology coverage consistently describe CL1 as containing approximately 800,000 neurons, the same order of magnitude as the original DishBrain prototype [1]. Cortical Labs' own Doom demonstration video, by contrast, states the specific unit shown contains "around 200,000" neurons [3]. Both statements come from the company itself; we could not find a reconciliation of the two figures in any source consulted for this paper, and we decline to silently pick one. Readers should treat CL1's neuron count as company-stated but internally inconsistent across the company's own communications, not as a settled specification.

## IV. Three Conditions for Replacement

A "yes" answer to this paper's title question requires all three of the following conditions to hold at once. Each is evaluated in turn.

### A. *Condition 1: Capability Parity*

Capability parity asks whether the biological substrate itself, not surrounding conventional hardware, is doing the learning a biocomputing system exhibits. We assess it, $C \in [0, 1]$, against four criteria: (i) whether the system demonstrates statistically significant, above-chance improvement on a closed-loop task; (ii) sample efficiency, i.e. how much interaction is required before that improvement appears; (iii) generalization, whether performance transfers to a task variant the system was not directly trained on; and (iv) the proportion of the pipeline's decision-making that is attributable to conventional computational scaffolding around the biological substrate, rather than to the substrate itself.

Criterion (iv) is the condition's decisive one, and the evidence for it is uncomfortable for a straightforward "yes." Independent commentary on the CL1 Doom demonstration reports a specific test: the system's developer, on suspecting a conventional AI component might be doing most of the work, replaced the neuron culture with pure random noise and found that learning stopped entirely [5]—indicating the culture was contributing something real, but that a conventional AI component was doing most of the surrounding decision work regardless. This is not grounded in a peer-reviewed source in this paper—the test is reported only in independent commentary on the demonstration, not in a company technical publication or peer-reviewed study, and we were unable to locate either—but it is a design-relevant caveat any capability score for these systems should carry, since it directly affects how much of an apparent capability advance is properly attributed to the biological component at all rather than to the silicon scaffolding this paper's title question is asking whether biology can replace. Verdict: capability parity is real but partial, and currently unverified at the substrate level for any system in this paper's case mapping.

### B. *Condition 2: Resource Superiority at Scale*

Resource superiority at scale asks whether the efficiency case that motivates the whole replacement comparison actually has a demonstrated path from a single cultured dish to anything resembling data-center throughput. Table I compares biocomputing against conventional silicon AI across the dimensions this condition requires.

The per-neuron efficiency case in Table I is real: brain-scale power draw against data-center-scale power draw, and human sample efficiency against machine sample efficiency, both favor biocomputing by a wide margin at the scale measured so far. But condition 2 asks about scale, not per-unit efficiency, and that is where the case runs into a concrete physical limit established in the peer-reviewed organoid literature—one no amount of funding or algorithmic cleverness touches. Without a vascular system to actively deliver oxygen and nutrients, living tissue can sustain viable cells only within a few hundred micrometers of its surface, because diffusion alone cannot supply an actively metabolizing core beyond that distance. Allometric modeling of organoid viability places the size ceiling at approximately 475 micrometers in radius for a gel-encased structure before a hypoxic, non-viable core begins to form [23]—a scale several orders of magnitude below anything resembling a data-center-scale processing unit. A biological brain avoids this limit through a dense capillary network that actively perfuses tissue rather than relying on passive diffusion; no biocomputing system studied in this paper, and none identified in the wider literature at time of writing, has demonstrated a comparable vascularization solution at any scale larger than a single small organoid. This is not a claim that biocomputing's efficiency advantage is fake—the per-neuron energy figures in Table I are real and independently sourced—it is a claim that the advantage does not currently have a demonstrated path to the scale at which it would need to operate to functionally replace, rather than supplement, conventional AI infrastructure. Claims to the contrary that circulate in press coverage—the "million times less power" figure this paper already flags as untraceable to any peer-reviewed FinalSpark publication, and larger, still less-sourced multipliers that appear in secondary technology-press aggregation rather than in any company or scientific publication this paper could verify—should be read against this unresolved scaling constraint, not independently of it. The only system-level operating-power figure located for a complete commercial unit—the ∼850–1,000 W drawn by a

TABLE I
BIOCOMPUTING VERSUS CONVENTIONAL SILICON AI. EFFICIENCY FIGURES ARE THE MOST DEFENSIBLE, PRIMARY-SOURCED VALUES FOUND DURING THIS PAPER'S LITERATURE VERIFICATION; SEVERAL WIDELY CIRCULATED FIGURES (10 GWH FOR GPT-3 TRAINING; "A MILLION TIMES" POWER ADVANTAGE FOR ORGANOID PLATFORMS) ARE MARKED AS UNVERIFIED BECAUSE THEY COULD NOT BE TRACED TO A PEER-REVIEWED PRIMARY SOURCE AT TIME OF WRITING.

| Dimension | Conventional AI | Biocomputing |
|---|---|---|
| Substrate | Silicon transistors (GPU/CPU/TPU dies) | Cultured human neurons on electrode arrays or in organoid form [2], [11] |
| Hardware state | Fixed architecture once fabricated; weights updated, circuit topology does not | Structurally plastic: connections between living neurons can physically reorganize during training, not only weighted [2] |
| Training energy (reference case) | 1,287 MWh, GPT-3 [18] | No comparably audited training-energy figure located; brain-scale estimate is ∼20 W continuous [20]. A full commercial CL1 rack, including life support, draws a company-communicated ∼850–1,000 W operating (not training) power—a launch-coverage figure, not a peer-reviewed one [22] |
| Compute/scale growth | ∼4.4×/yr since 2010; 3–6 month doubling for frontier models [19] | No comparable trend data published; single generations (Pong→Doom) improved over ∼4 years [3] |
| Learning mechanism | Gradient descent over a fixed, differentiable objective | Predict-and-minimize-surprise stimulus/response shaping, not backpropagation [2] |
| Maturity | Multi-decade, standardized tooling | Single-digit years; two companies with materially different architectures [11] |
| Moral-status risk (Condition 3) | Effectively zero under current consciousness science | Nonzero and currently unmeasurable (Section IV-C) |

full CL1 rack, life-support subsystem included [22]—cuts the other way from the "million times less" framing: it is a real device drawing power in the same order of magnitude as a single high-end GPU, not a negligible fraction of one, once the non-neural infrastructure a cultured-neuron system actually requires to stay alive is counted rather than quietly excluded.

**Key Finding —** Biocomputing's efficiency advantage is real per neuron but has no demonstrated path past a ∼475-micrometer diffusion-limited size ceiling—nowhere near the scale data-center deployment would require—making condition 2 the clearest of the three conditions to fail, and the one least likely to be resolved by funding or algorithmic progress alone.

Verdict: resource superiority at scale fails, not on economic or algorithmic grounds, but on an unsolved bioengineering constraint—vascularizing cultured neural tissue—that no system in this paper's case mapping has published a solution to.

### C. Condition 3: No Disqualifying Cost

Even where conditions 1 and 2 might eventually be satisfied, a genuine replacement claim requires that pursuing it carry no cost conventional silicon AI is exempt from. This paper argues biocomputing fails this condition by construction, because capability and moral-status risk move along separate axes entirely—pushing one upward does nothing to the other, in either direction.

We define moral-status risk, $R \in [0, 1]$, deliberately *not* as an estimated probability of consciousness; no instrument exists that could produce such an estimate, and this paper does not claim otherwise. Instead we define $R$ as an enumerable checklist score across five substrate properties that current consciousness science treats as relevant, whether or not any of them is individually sufficient: (i) living versus non-living tissue; (ii) presence of a homeostatic, predict-and-minimize-surprise feedback loop of the kind Solms associates with the origin of affect [13]; (iii) structural organization, with a flat two-dimensional culture scoring lower than a three-dimensional organoid with self-organized layered structure; (iv) human, as opposed to non-human animal or fully synthetic, cellular provenance; and (v) absence of a validated test capable of ruling experience out, which we include as a criterion in its own right because Owen et al.'s finding demonstrates that absence of behavioral evidence is not evidence of absence [14].

The claim that $C$ and $R$ are independent, not two ends of a single dial, is a design argument, not an empirical one, and it can be shown by construction rather than by appeal to any single system's measured values.

*Claim.* There exist, or could straightforwardly exist, systems $s_1, s_2$ such that $C(s_1) < C(s_2)$ while $R(s_1) = R(s_2)$, and systems $s_3, s_4$ such that $R(s_3) < R(s_4)$ while $C(s_3) = C(s_4)$.

*Argument for the first pair.* DishBrain and CL1 running its early Pong-only configuration versus CL1 running its matured Doom configuration satisfy this directly. Section III establishes that the interface and encoding scheme improved substantially between the two—longer training time and a simpler task in one case, a harder task solved faster with a matured API in the other—which is precisely what criterion (ii) and (iii) of $C$ are built to register as a difference. None of $R$'s five criteria changed: both are living, human-derived, cultured on a flat electrode array, trained with the same predict-and-minimize-surprise loop, and neither can be ruled out or in on experience by any existing test. A capability advance occurred with the risk profile held fixed.

*Argument for the second pair.* Consider a flat two-dimensional neuron culture that has not yet been embodied in any closed-loop task at all—$C \approx 0$ by criterion (i), since no learning has been demonstrated—against a fully-formed, three-dimensionally organized organoid that has likewise not been embodied in any task. Both score $C \approx 0$. But the organoid scores higher on $R$ purely by virtue of criterion (iii)'s structural-organization term: a self-organized, layered, three-dimensional structure is closer to features consciousness

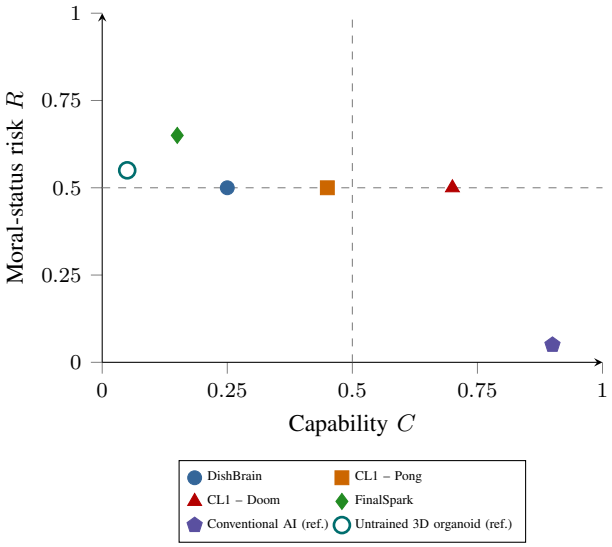



Fig. 2. The four case-mapped systems from Table II plotted against both axes, with a conventional-AI reference point (high capability, near-zero risk) and an untrained 3D organoid reference point (near-zero capability, nonzero risk) illustrating Section IV-C's non-collapsibility argument. Positions are qualitative placements within the bands of Table II, not measured scores.

science treats as potentially relevant than an unstructured flat sheet is, independent of whether either has ever been asked to do anything. Risk, on this account, can be present before capability is demonstrated at all, which is exactly why treating an untrained system as risk-free because it has not yet "done anything impressive" is a category error, not merely an overcautious position.

**Key Finding —** Capability and moral-status risk can move independently in both directions: a capability jump can occur with risk held fixed (DishBrain to CL1-Doom), and risk can be nonzero before any capability is demonstrated at all (an untrained three-dimensional organoid). Every efficiency gain condition 2 might eventually deliver is therefore bought with a risk category conventional silicon AI is exempt from entirely—the disqualifying cost condition 3 asks about.

A corollary follows directly: because $R$ can be nonzero at $C \approx 0$, a governance or reporting policy that gates ethical scrutiny on capability milestones—review it once it does something impressive—is misaligned with what the checklist criteria above actually track. The living-tissue and human-provenance criteria are present at the moment cells are cultured, not at the moment a demonstration succeeds. Verdict: condition 3 fails by construction, independent of how conditions 1 and 2 eventually resolve.

## V. Case Mapping

Table II applies conditions 1 and 3 to the three systems this paper has discussed in detail, plus the DishBrain-versus-CL1-Doom contrast that Section IV-C's theoretical argument depends on directly; Fig. 2 plots the same systems spatially, alongside the two reference points used in that section's construction argument.

TABLE II
CASE MAPPING ACROSS CAPABILITY AND MORAL-STATUS RISK. $C$ AND $R$ ARE QUALITATIVE BANDS (LOW/MODERATE/HIGH), NOT THE NUMERIC SCORES SECTION IV DEFINES, SINCE NO SYSTEM HERE HAS BEEN SCORED AGAINST A VALIDATED INSTRUMENT FOR EITHER AXIS.

| System | Capability | Risk | Notes |
|---|---|---|---|
| DishBrain (2022) | Low–Moderate | Moderate | First closed-loop demonstration; flat culture, human+mouse mix [2] |
| CL1 – Pong config. | Moderate | Moderate | Same risk profile as DishBrain; slower task [1] |
| CL1 – Doom config. | Moderate–High | Moderate | Harder task, faster solve; risk profile unchanged from Pong config. [3] |
| FinalSpark Neuroplatform | Low (documented) | Moderate–High | Peer-reviewed paper makes no game-learning claim; 3D organoid structure scores higher on criterion (iii) [11] |

The pattern Table II is built to surface is that capability varies substantially across these four rows while risk does not vary nearly as much—the opposite of what a single collapsed scale would predict, where the two columns would move together. FinalSpark's entry is the clearest illustration: its peer-reviewed system paper documents no closed-loop game-learning result comparable to DishBrain or CL1, so its capability entry is markedly lower, yet its use of genuine three-dimensional organoids, rather than flat cultures, means it does not obviously score lower on risk [11]. None of the four rows satisfies all three of Section IV's conditions simultaneously, which is exactly the paper's title question answered at the level of individual systems rather than the category as a whole.

The same separation applies directly to embodiment, a direction none of the three systems mapped above have taken but that this evaluation already covers without modification. A robot whose motor output is driven by a conventional silicon controller scores near-zero on every one of $R$'s five criteria regardless of how behaviorally sophisticated its controller becomes, since none of living tissue, affect-associated feedback, structural organization, human provenance, or undetectability by existing tests applies to a non-biological substrate—capability and risk stay decoupled in exactly the direction Section IV-C's first construction argument describes. A robot whose motor output is instead driven by a cultured neuron population inherits that population's risk profile unchanged by the fact that its actions move a physical chassis rather than a simulated paddle or Doom avatar: what the motor output controls is a capability-axis question, specifically criterion (iii)'s task complexity, not a risk-axis one. A more behaviorally capable biocomputing-driven robot is not, on that basis alone, a higher-risk one; a biocomputing-driven robot of any capability level is not, on that basis alone, a lower-risk one than the dish-bound systems in Table II—and would still face condition 2's scaling ceiling if deployed at any consequential number.

## VI. Limitations

This evaluation has five limitations worth naming precisely rather than gesturing at generally.

The most consequential limitation is that condition 3's risk axis has no validated measurement instrument, and this paper does not supply one. The five checklist criteria in Section IV-C are motivated by consciousness science but are not themselves a scored, validated test; two different researchers applying the checklist to the same system could reasonably disagree on borderline items, most plausibly criterion (ii)'s judgment of whether a given feedback loop counts as sufficiently affect-like. That is not a flaw specific to this paper's execution; it reflects that no such instrument exists anywhere in the consciousness-science literature this paper draws on, and building one is explicitly future work in Section VII, not something resolved here.

Second, condition 1's fourth criterion—how much of a system's apparent intelligence is attributable to conventional computational scaffolding rather than the biological substrate—currently rests on a single documented test result on a single system (the noise-substitution finding on CL1's Doom demonstration) rather than a systematic study across multiple systems and task types. Generalizing that one result into a standing criterion is a reasonable design choice, not an established empirical pattern, and it means the paper's capability-parity verdict is more tentative than its resource-superiority and disqualifying-cost verdicts.

Third, the case mapping in Table II uses qualitative bands rather than the numeric $[0, 1]$ scores Section IV defines, because no system studied here has actually been scored against a validated instrument for either axis—the same gap named as the first limitation, surfacing again as a concrete measurement problem rather than an abstract one.

Fourth, this paper's own comparison in Table I is itself asymmetric in a way worth naming: conventional AI's efficiency figures are backed by audited, peer-reviewed primary sources, while several of the most dramatic biocomputing efficiency claims circulating in press coverage could not be traced to any peer-reviewed source during this paper's literature verification and are explicitly marked unverified rather than included as fact. That asymmetry may reflect real difference in how mature the reporting infrastructure is for each technology rather than a real difference in the underlying efficiency claims themselves, and this paper cannot distinguish between those two explanations.

Fifth, this evaluation is scoped to biocomputing systems built from cultured neurons on electrode arrays or organoids. Whether the same three-condition logic usefully extends to other substrates sometimes discussed alongside biocomputing—engineered brain-machine interfaces built from living tissue implanted in an existing organism, for instance—is a question this paper does not attempt to answer, since none of the systems studied here involve an intact nervous system, and any such system would need its own condition-2 scaling analysis rather than inheriting this paper's vascularization ceiling by assumption.

## VII. Conclusion

This paper's title question has a direct answer: no, not currently, and not for diffuse or merely economic reasons. Capability parity (condition 1) is real but partial—at least one system's apparent intelligence traces substantially to conventional silicon scaffolding around the neurons, not the biological substrate alone. Resource superiority at scale (condition 2) fails on an unsolved bioengineering limit documented in the organoid literature: cultured neural tissue cannot sustain a viable core beyond roughly 475 micrometers in radius without vasculature no biocomputing system has demonstrated, a ceiling nowhere near what data-center-scale deployment would require. No disqualifying cost (condition 3) fails by construction: capability and moral-status risk vary independently of one another, so every efficiency gain biocomputing might eventually deliver is bought with a risk category conventional silicon AI does not carry at all.

Future work is not symmetric across the five limitations named in Section VI, and treating it as five equally weighted items would misstate how urgent each gap actually is. Building a validated instrument for condition 3's risk axis is the single highest-priority item, because every other use of this evaluation—the case mapping, the comparison table, any future governance policy built on top of either—inherits that gap without it: a checklist score that cannot be validated against anything is a structured opinion, not yet a measurement. Systematically testing condition 1's scaffolding-attribution criterion across more than one system and task type is the second priority, since Section VI already flagged that criterion as resting on a single result, and the paper's capability-parity verdict is only as strong as that evidence. Whether condition 2's vascularization ceiling is solvable at all—and on what timescale, if so—is a real, empirically answerable engineering question, but it is the least urgent of the three to resolve first, since a positive answer to conditions 1 and 3 without it still leaves replacement physically unrealized, while a negative answer to either of the other two makes the scaling question moot regardless of how condition 2 eventually resolves.

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